Inventory product management
Ideation Phase
Brainstorm & Idea Prioritization

Date	07/03/2026
Team ID	SWTID-2026-5599
Project Name	Inventory Product Mangement
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template

Brainstorming is used to generate creative ideas and identify the most suitable solution for a real-world problem. During the ideation phase, team members collaborated to discuss common issues faced in managing product inventories and explored multiple web-based solutions using the MERN stack. Ideas were evaluated based on feasibility, scalability, security, and learning outcomes.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

The team conducted a discussion session to identify common problems that could be solved using full-stack development. Several ideas were proposed such as:

E-commerce product tracker
Expense manager
Task management system
Cryptocurrency dashboard
Inventory management system

After evaluating complexity, real-world usefulness, and alignment with MERN stack learning objectives, the team finalized the following problem statement:

Problem Statement:
Small businesses and individual users lack a simple, secure, and centralized system to manage their product inventory. Many existing solutions are complex or require paid subscriptions. Therefore, the team decided to build a **JWT-based Product Inventory Management System** that allows users to securely manage their own products.

Step-2: Brainstorm, Idea Listing and Grouping

During brainstorming, all team members contributed ideas without restrictions. The ideas were then grouped into categories:

Authentication & Security Ideas

JWT-based login system
User-specific product access
Secure password hashing
Role-based access (future scope)

Core Inventory Features

Add product
Update product details
Delete product
View all products
Search products by name

User Experience Ideas

Simple dashboard UI
Responsive design
Clean product cards layout
Fast loading using React

Technical Ideas

MERN stack architecture

REST API development
MongoDB database design
Middleware for route protection

After grouping, the team focused on ideas that were realistic within the project timeline.

Step-3: Idea Prioritization

The team prioritized ideas based on:

Feasibility within deadline
Learning value
Practical real-world use
Technical complexity

High Priority (Implemented Features)

JWT Authentication (Register/Login)
Private product CRUD operations
MongoDB database integration
Secure API routes
Search functionality

Medium Priority (Enhancements)

Product filtering
Improved UI components
Error handling improvements

Low Priority / Future Scope

Role-based admin system
Analytics dashboard
Notifications for low stock
Deployment and cloud hosting

Final Decision:
The team finalized the **Product Inventory Management System** as it provides strong learning opportunities in authentication, backend API design, and frontend integration using MERN stack technologies